

Final Project Report

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Subject

File Metadata Analysis in Digital Forensics

The following research examines how investigators extract and analyze metadata to uncover digital crimes. Metadata essentially means “data about data” and metadata of a file contains valuable forensic evidence as it provides a layer of information apart from the content itself. The main goal of computer forensics is to reveal digital evidence and use it in legal proceedings which can prove or disprove illegal activity. Metadata is an important piece of digital information to be collected and analyzed upon data retrieval on computers, hard drives, and other digital media.

Topics

Metadata

The following describes what metadata reveals about digital evidence. Three types of metadata exist to reveal important information of data: descriptive, structural, and administrative. Descriptive metadata show the content of the data that show the relevance and purpose of the data. Structural metadata shows the relationships and connections that the data may have. Administrative metadata shows technical information such as the creation date, file author, and file format. A combination of the different types of metadata reinforces the credibility and completeness of the relevant digital evidence and extends upon the content of the data. Each type of metadata can show when a file was created, modified, accessed, by whom, and where it was created or captured. Metadata can help investigators or forensic experts to confirm the authenticity of data or spot tampering within it.

Metadata essentially acts as a digital fingerprint that authenticates and validates the digital evidence by providing the evolution, context, timestamps, and origin of the evidence. Metadata is especially used in the chain of custody process to uphold the credibility of the evidence by recording timestamps, locations, and user interactions of the evidence, serving as a counterpart to physical seals and signatures. It can show how evidence was acquired, preserved, and presented which introduces us to the next topic.

Techniques

Extraction

Metadata extraction refers to the retrieval and collection of metadata from files that can be found on computers, storage devices, and networks. Common techniques to extract metadata involve disk imaging and file system forensics. Disk imaging involves creating an exact copy of storage devices that includes all its metadata and captures the exact state of the device at the time. File system forensics analyzes the file system of storage devices that include metadata about files and directories which can be used to reconstruct the file system and identify relevant evidence. After retrieving the data along with the metadata, we can use a variety of tools to view and analyze the metadata.

The techniques for extracting metadata use a variety specialized programs, online services, or built-in tools that depend on the file type and depth required in the file analysis. A couple of examples to extract metadata are ExifTool (command-line) for a huge number of image formats and MediaInfo for video and audio files. For documents, programs such as Microsoft Word have built-in property viewers that show basic information about the document such as the author, creation data, and editor.

ExifTool is an open-source command-line tool that is one of the most notable tool of them all. It was created by Phil Harvey and the tool supports metadata extraction for approximately 180 file formats which is highly important in computer forensic investigations. To use the tool in the command line, you must run “exiftool filename.ext” to view the metadata for only a single file which appears in the terminal. There are many tags you can add as additional options to the command such as “-a” and “-s” for duplicates and short tag names. You can also add the “-r” recursion option to scan subdirectories of a folder.

Preservation

Knowing the best practices for metadata preservation is important because small key details can prove the relevance and important of digital information. The following paragraphs explain the best practices to preserve metadata, how it the methods work, and why it is effective.

One of the most important best practices is to essentially create a backup and reserve an original, unchanged set of files and only process from copies. You risk losing information if you are processing information and accidentally delete or change a file which can be difficult to revert or replicate the same file if you do not have a backup. You should also not assume what metadata is included in a file to consider whether it is safe to share with another party unless absolutely required. Metadata could contain basic information such as date and time created, but it can also include privileged information that should not be shared. Therefore, you should also not assume that there is privileged information and confirm what exactly is included in the metadata and communicate whether the data needs to be withheld.

On terms of handling files, you should not convert documents into certain file formats such as PDF or TIFF because it could alter the content and strip away metadata such as creation data. Therefore, do not use TIFF and PDF as an acceptable format. For spreadsheet files, you should check for formulas and hidden cells to keep document integrity.

Lastly, you should document the necessity of metadata and discuss about it as soon as possible. Without showing why the metadata is needed, the court may not grant the request for metadata collection. Metadata should be included in discussions at early stages of legal proceedings to ensure the preparation of the trial runs smoothly and efficiency by understanding the required information is available when it is needed.

Challenges

The first challenge in counter-forensics is full-disk or file-based encryption technologies that make it more difficult to extract metadata because it can require decryption keys that are hard to access without cooperation of the original data owner or service provider. Additionally, malicious actors can use anti-forensic techniques in attempts to hide metadata from extraction or destroy the digital evidence entirely.

The volume and complexity of the data on modern digital devices can provide a challenge in metadata extraction as large storage capacities, various file formats, and complex data structures may require expensive tools and techniques to adequately extract and analyze important information. Data corruption can also occur from file system errors, disk damage, or attempts to hide evidence where special tools are needed to recover and reconstruct digital information. Metadata can be also found on cloud and remote environments stored in multiple locations and jurisdictions which may require extensive cooperation from service providers and compliance to legal requirements.

Lastly, metadata extraction may be affected by hardware and software compatibility, access restrictions, privacy and legal limitations. Extracting metadata from a proprietary or legacy system may require a specific tool or expertise for that device or environment.

References

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